

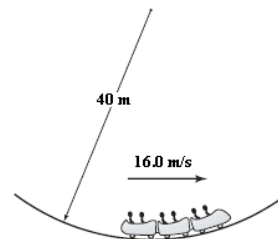
2026 PH1012 Tut 5: Circular Motion

Horizontal Circular Motion

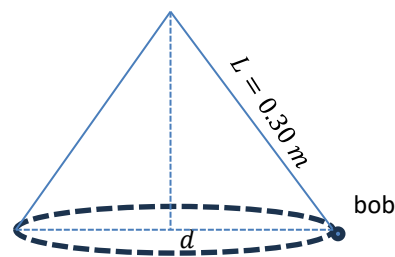
1. (Class Activity 1) Exploring the effects of forces / accelerations on velocities [See if you can draw a circle, parabolic path]: <https://www.glowscript.org/#/user/matterandinteractions/folder/matterandinteractions/program/02-Newton-grid>
2. [G5.51] [*] A coin is placed 12.0 cm from the axis of a rotating turntable of variable speed. When the speed of the turntable is slowly increased, the coin remains fixed on the turntable until a rate of 35.0 rpm (revolutions per minute) is reached, at which point the coin slides off. What is the coefficient of static friction between the coin and the turntable?

18 min Vertical Circular Motion / Conical-pendulum type of Motion (Summary + Q 3-5)

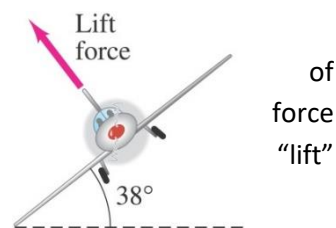
3. [*] A roller-coaster car has a mass of 500 kg when fully loaded with passengers. At the bottom of a circular dip of radius 40 m (as shown in the figure) the car has a speed of 16 m/s. What is the magnitude of the force of the track on the car at the bottom of the dip?
4. [*, Claude generated] The same fully loaded roller-coaster car of mass 500 kg now travels around the inside of a vertical circular loop of radius 10 m. At the top of the loop, its speed is 14 m/s. (a) What is the magnitude of the force of the track on the car at the top of the loop? (b) What is the minimum speed at the top for the car to maintain contact with the track?
5. [G5.53] [*] A 975-kg sports car (with driver) crosses the rounded top of a hill of radius 88.0 m at 12.0 m/s. Determine (a) the normal force exerted by the road on the car, (b) the normal force exerted by the car on the 72.0-kg driver, and (c) the car speed at which the normal force on the driver equals zero.



6. 18 min [Class activity] In this activity, you are challenged to make a conical pendulum with a period of $T = 1$ s using an inextensible string of 0.30 m and a mass.
 - (a) By analyzing the forces and applying kinematic equations, determine the diameter of the circular motion of the mass so that the period of pendulum is 1 s.
 - (b) Make the pendulum swing so that the diameter remains as the value as determined above.
 - (c) Measure the period and check if it agrees with theoretical calculation.



7. 10 min [G5.91] [**] An airplane traveling at 480 km / h needs to reverse its course. The pilot decides to accomplish this by banking the wings at an angle 38° . (a) Find the time needed to reverse course. (b) Describe any additional the passengers experience during the turn. [Hint: Assume an aerodynamic force that acts perpendicularly to the flat wings; see Figure.]

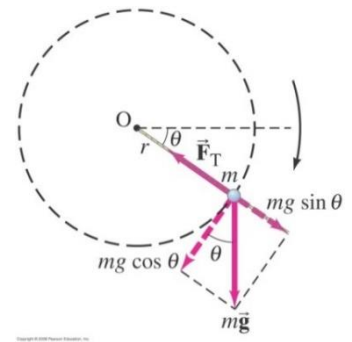


Angular Acceleration

(Class Activity 2) Exploring the effects of forces / accelerations on velocities [See if you can trace out a path of an object moving in a circular path with increasing speed]:

<https://www.glowscript.org/#/user/matterandinteractions/folder/matterandinteractions/program/02-Newton-grid>

8. [G10.12] [*] A 64-cm-diameter wheel accelerates uniformly about its center from 130 rpm to 280 rpm in 4.0 s. Determine (a) its angular acceleration, and (b) the radial and tangential components of the linear acceleration of a point on the edge of the wheel 2.0 s after it has started accelerating. [Video]
9. [G10.21] [**] A small rubber wheel is used to drive a large pottery wheel. The two wheels are mounted so that their circular edges touch. The small wheel has a radius of 5.5 cm and accelerates at the rate of 7.5 rad/s^2 , and it is in contact with the pottery wheel (radius 25.0 cm) without slipping. (a) Calculate the angular acceleration of the pottery wheel. (b) Calculate the time it takes the pottery wheel to reach its required speed of 65 rpm. [Video]
10. 10 min [G5.100] [*] A ball of mass $m = 1.0 \text{ kg}$ at the end of a thin cord of length $r = 0.80 \text{ m}$ revolves in a vertical circle about point O, as shown in the figure. During the time we observe it, the only forces acting on the ball are gravity and the tension in the cord. The motion is circular but not uniform because of the force of gravity. The ball increases in speed as it descends and slows down as it rises on the other side of the circle. At the moment the cord makes an angle $\theta = 30^\circ$ below the horizontal, the ball's speed is 6.0 m/s . At this point, determine the tangential acceleration, the radial acceleration, and the tension in the cord F_T . Take θ increasing downward as shown.



Past Mid-term / Exam question will only be released during the tutorial.

Answers

1. -
2. $\mu_s = 0.164$
3. 8100 N
4. 4900 N; 9.9 m/s
5. (a) 7960 N; (b) 588 N; (c) $v = 29.4 \text{ m/s}$
6. -
7. 55s, Normal force increase to 1.27 mg .
8. a) 3.9 rad/s^2 ; b) 148 m/s^2 , 1.4 m/s^2
9. $\alpha_p = 1.65 \text{ rad/s}^2$ (b) $t = 4.13 \text{ s}$
10. $a_{\text{tan}} = 8.5 \frac{\text{m}}{\text{s}^2}$; $a_{\text{rad}} = 45 \frac{\text{m}}{\text{s}^2}$; $F_T = 50 \text{ N}$